

DOUGLAS MENDES BARRETO

Senior Software Engineering Manager

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Professional Summary

Hands-on engineering leader with **14+ years** of experience building products, platforms, and engineering teams. I currently lead a cross-functional engineering team while still contributing in code when it helps unblock delivery, improve developer experience, or solve customer-facing issues.

My leadership style is grounded in high ownership, pragmatic execution, and strong technical judgment. I do my best work in environments that trust engineers with broad responsibility, value written communication, and invest in leverage through tooling, abstractions, and thoughtful team design.

Highlights for Ashby

- Led a **9-engineer** team spanning product, platform, and delivery while contributing hands-on to backend and frontend systems
 - Built a self-service AI prototyping workflow that reduced proof-of-concept delivery time from **2 days to 4 hours**
 - Rolled out AI tooling and training across engineering, increasing per-engineer productivity by **3x** while maintaining quality
 - Re-architected CI/CD pipelines, reducing build and deploy time from **50 minutes to 7 minutes**
 - Increased delivery velocity by **5x**, cutting lead time from **13 weeks to 2 weeks**
 - Introduced engineering standards and collaboration practices that reduced bug rates by **23%**
 - Led cloud migration from AWS to Google Cloud, reducing infrastructure costs by **20%**
 - Improved product and engineering collaboration by reducing planning overhead and unnecessary meetings
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Management Skills

- Coaching engineers to independently scope, own, and deliver loosely defined projects
- Engineering leadership across product, platform, and developer experience initiatives
- Hiring support, performance management, mentorship, and growth planning across seniority levels
- Strong technical feedback on architecture, implementation approaches, delivery plans, and quality tradeoffs
- Async-first communication through clear proposals, documentation, and stakeholder alignment
- Building team leverage through workflow automation, CI/CD improvements, shared tooling, and platform investments
- Pragmatic delivery leadership using Kanban, trunk-based development, and lightweight process
- Hands-on debugging and implementation across backend, frontend, and mobile systems

Tech Skills

- **Languages:** Python, JavaScript/TypeScript, Go, C++, C#, Ruby, Swift, Objective-C
 - **Product & Platform:** Next.js, React Native, Node.js, GraphQL, REST APIs, microservices, event-driven systems
 - **Cloud & DevOps:** Google Cloud, Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, CI/CD
 - **Data:** MySQL, PostgreSQL, MongoDB, Redis
 - **Developer Productivity:** GitHub Copilot, Claude Code, Codex, workflow automation, AI-assisted prototyping
 - **Quality & Delivery:** Unit, integration, and E2E testing, TDD/BDD, trunk-based development, Kanban
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Professional Experience

Senior Software Engineering Manager

Xpertsea - Predictive shrimp biomass control with AI delivering sustainable aquaculture solutions

51 - 100 employees

Quebec, Canada

Sep 2020 – Present

- Lead a 9-engineer team responsible for product, platform, and delivery across multiple initiatives
- Built a self-service AI prototyping workflow that reduced PoC delivery time from 2 days to 4 hours
- Deployed AI tooling, team training, and repeatable operating practices, increasing per-engineer productivity by 3x while maintaining quality
- Coached engineers across execution, technical decision-making, and cross-functional collaboration
- Re-architected CI/CD pipelines, cutting build and deploy time from 50 minutes to 7 minutes
- Increased feature delivery velocity by 5x, reducing lead time from 13 weeks to 2 weeks
- Introduced trunk-based development to reduce overhead and improve flow
- Implemented Kanban practices that reduced planning overhead by 20%
- Redesigned engineering and product collaboration workflows to reduce communication friction and unnecessary meetings
- Introduced engineering standards that reduced bug rates by 23% and improved async collaboration
- Led platform re-architecture efforts that improved developer experience, user experience, and product scalability
- Spearheaded a full-stack migration from AWS to Google Cloud, reducing infrastructure costs by 20%
- Oversaw the launch of 6 major projects across the platform
- Managed people operations including hiring support, performance reviews, mentorship, and conflict resolution
- Contributed hands-on code to backend systems (Python, TypeScript) and frontend applications (Next.js)

- Built a cross-platform C++ computer vision framework for native iOS and Android applications
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Senior iOS Developer

Social.mom - Social media platform for mothers focused on delivering a safe and supportive environment

11 - 20 employees

Sep 2019 – Sep 2020

- Engineered and maintained a high-traffic iOS application while leading migration of key features from Objective-C to Swift
 - Led a major re-architecture of the native codebase, reducing maintenance overhead and cutting feature development time by 20%
 - Built automation scripts that eliminated hundreds of hours of repetitive manual work for the team
 - Established unit and E2E testing practices to improve reliability and release confidence
 - Mentored engineers on software design, SOLID principles, and modern architecture patterns
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Senior Software Engineer

Levio - Tech Consulting

1K - 5K employees

Jun 2018 – Sep 2019

- Engineered and maintained scalable, high-availability backend services for multiple clients using .NET C#
 - Delivered legacy modernization projects, migrating critical systems from COBOL to modern .NET platforms
 - Led migration from legacy version control systems to Git, improving collaboration and source control practices
 - Worked closely with cross-functional teams to define requirements and deliver robust solutions
 - Improved CI pipelines through optimization and automation to support faster delivery
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Senior Mobile Engineer

Concrete Solutions - Tech Consulting

10K+ employees

Jun 2015 – Jul 2018

- Progressed from Developer to Senior Engineer, leading architecture and development for native mobile applications across multiple clients
- Delivered mobile banking and e-commerce products in regulated, high-scale environments
- Architected and developed a white-label e-commerce mobile platform for rapid rollout from one codebase

- Delivered internal technical talks on coding standards, testing practices, and QA implementation
 - Designed and implemented CI/CD pipelines while promoting TDD/BDD practices
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Lead Game Programmer

Editora Globo - Publisher and developer of digital entertainment products

500 - 1K employees

Jul 2010 – Mar 2015

- Progressed from Programmer to Lead, combining people leadership with hands-on technical ownership
 - Led end-to-end development of cross-platform games for iOS, Android, and Windows using C# and Unity3D
 - Architected and built backend services for MMO games using Java and MySQL
 - Designed reusable engine tooling and framework components that improved team productivity
 - Implemented JIRA to formalize planning and execution processes
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Junior Analyst Programmer

People Way - Tech Consulting

11 - 50 employees

Mar 2009 – Jan 2010

- Developed and maintained enterprise web applications using J2EE, JSP, JSF, and SOAP web services
 - Managed client data lifecycle work including MySQL database design and JasperReport-based reporting
 - Built data parsing scripts to process SMS, XML, and CSV sources
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Education

University Anhembi Morumbi

Bachelor's Degree in Game Design and Development - 2010

Languages

- English: Fluent
- Portuguese: Native
- French: Professional

Selected Projects

Kampi (Xpertsea)

Intelligent copilot for shrimp farmers using AI and computer vision to replace manual guesswork with data-driven insights. Features three pillars: Pond Vision for biometric monitoring via smartphone photos, Optima for predictive analytics and growth forecasting, and FarmBook for team coordination across the production chain.

Tech Stack: Python, TypeScript/JS, MongoDB, MySQL, Node.js, Bun, Next.js, Google Cloud, Terraform, Ansible, C++, Android, iOS, React Native + Native Development

Link: <https://bekampi.com/>

- Led cross-functional engineering work across AI-powered product, platform, and operations systems
 - Defined technical direction for predictive systems and computer vision capabilities
 - Managed technical roadmap for an omnichannel platform supporting iOS, iPadOS, and web
 - Directed infrastructure modernization using Google Cloud, Terraform, and Ansible
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Social.Mom (Social.mom)

Social network platform designed to counter social isolation during motherhood and reinforce the social safety net of families. Enables mothers to connect locally, plan activities, and access community groups and organizations in their area through a bilingual interface.

Tech Stack: iOS native, Android native, Java Spring Boot, React

Link: <https://social.mom/en>

- Engineered and maintained a high-traffic iOS application with bilingual support in English and French
 - Led re-architecture work that reduced maintenance overhead and accelerated feature development by 20%
 - Established formal testing practices with unit and E2E coverage for better reliability
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Mundo do Sítio (Editora Globo)

Persistent, multi-user virtual world designed to integrate pedagogical goals with immersive gameplay as a digital extension of the Sítio do Picapau Amarelo IP. Featured a complex economy, social interactions, and a library of educational mini-games for millions of Brazilian students.

Tech Stack: Unity 3D, Flash ActionScript (SWC), Java, SmartFox Server, HTML5/JS

- Engineered full-stack architecture for high-concurrency traffic and low-latency communication
- Developed gameplay systems and real-time state synchronization for 30+ educational mini-games
- Implemented secure authentication, profile registration, and parental controls for a children's platform